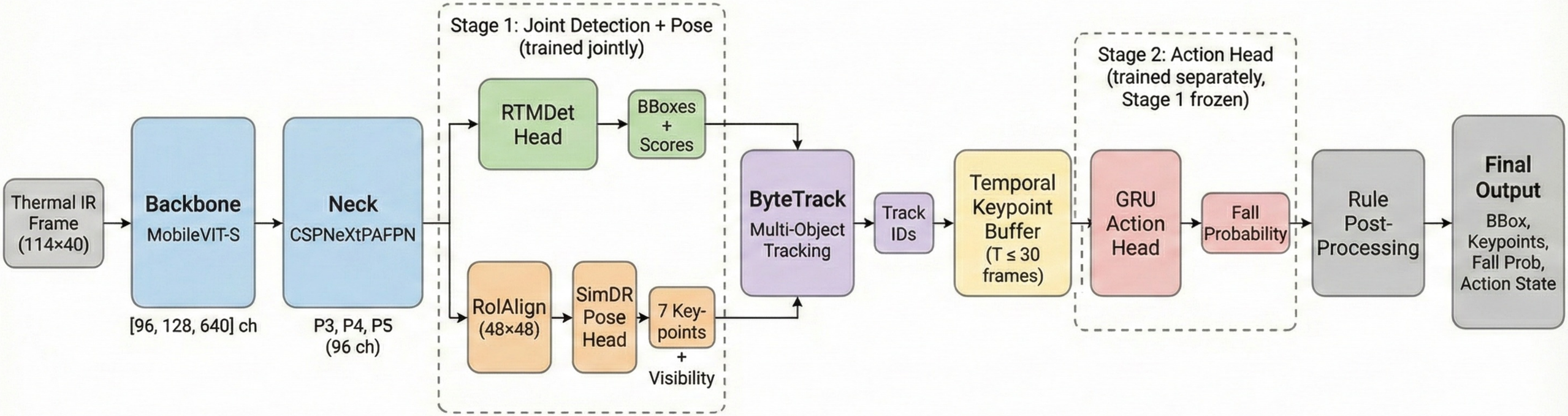
Thermal IR Frame (114×40)
Backbone
MobileVIT-S
[96, 128, 640] ch
Neck
CSPNeXtPAFPN
P3, P4, P5 (96 ch)
Stage 1: Joint Detection + Pose (trained jointly)
RTMDet Head
BBoxes + Scores
RoIAlign (48×48)
SimDR Pose Head
7 Key-points + Visibility
ByteTrack
Multi-Object Tracking
Track IDs
Temporal Keypoint Buffer (T ≤ 30 frames)
Stage 2: Action Head (trained separately, Stage 1 frozen)
GRU Action Head
Fall Probability
Rule Post-Processing
Final Output
BBox, Keypoints, Fall Prob, Action State